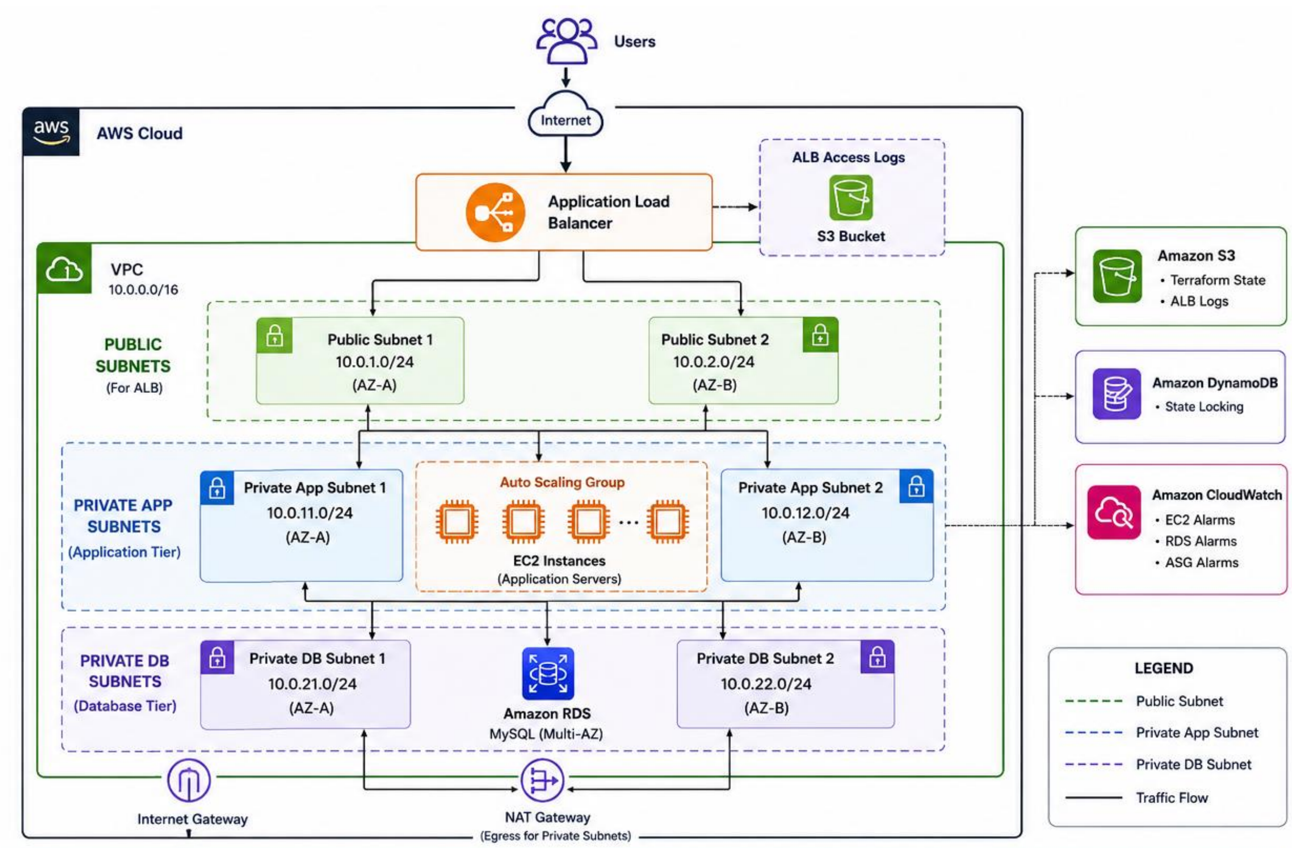
Production-Grade 3-Tier AWS Infrastructure using Terraform

Overview

This project demonstrates the deployment of a scalable, highly available, and secure 3-tier application architecture on AWS using Terraform.

The infrastructure was initially developed as a monolithic Terraform configuration and later refactored into reusable modules following Infrastructure as Code best practices.

Topology



Architecture

Presentation Layer:

- Application Load Balancer (ALB)

Application Layer:

- EC2 Instances
- Launch Template
- Auto Scaling Group

Database Layer:

- Amazon RDS MySQL

Networking:

- VPC
- Public Subnets
- Private Application Subnets
- Private Database Subnets
- Internet Gateway
- NAT Gateway
- Route Tables

Monitoring and Logging:

- CloudWatch Alarms
- ALB Access Logs

State Management:

- Amazon S3 Backend
- DynamoDB State Locking

Objectives

- Implement Infrastructure as Code using Terraform
- Design a secure 3-tier architecture
- Achieve high availability across multiple Availability Zones
- Implement monitoring and logging
- Manage Terraform state remotely
- Apply modular Terraform design principles

Technologies Used

- Terraform
- AWS EC2
- AWS VPC
- AWS ALB
- AWS Auto Scaling
- AWS RDS
- AWS CloudWatch
- AWS S3
- AWS DynamoDB
- GitHub

Project Phases

Phase 1: Networking Setup Phase 2: Security Groups Phase 3: Application Load Balancer Phase 4: Launch Template and Auto Scaling Phase 5: Amazon RDS Phase 6: Outputs and Testing Phase 7: CloudWatch Monitoring Phase 8: Auto Scaling Policies Phase 9: Remote State Management Phase 10: Terraform Modularization Phase 11: ALB Access Logging Phase 12: Project Validation and Optimization

Key Learnings

- Terraform state management
- Remote backend implementation
- Infrastructure modularization
- AWS networking fundamentals
- Load balancing and high availability
- Auto Scaling strategies
- Monitoring and observability
- Production infrastructure design

Future Enhancements

- GitHub Actions CI/CD
- HTTPS using ACM
- Route53 Custom Domain
- AWS WAF
- Multi-Environment Deployments
- Terraform Workspaces
- Blue/Green Deployments